

Activity: Retail Analytics

Scenario:

You are working as a **Data Analyst** in a company that supplies products across multiple markets and regions.

The company sells products across:

- Multiple **Categories**
- Different **SubCategories**
- Various **Customer Segments**
- Multiple **Regions**

Management currently struggles because:

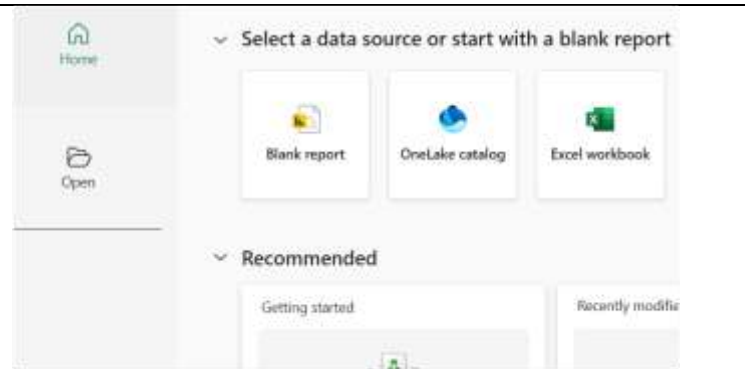
- They cannot clearly see revenue contribution by product hierarchy.
- They cannot track sales growth trends properly.
- They do not know if they are meeting annual targets.
- They cannot easily identify which region or segment drives profit.

The company has now implemented a **Self-Service BI Tool**, and your job is to design a dashboard that answers key business questions.

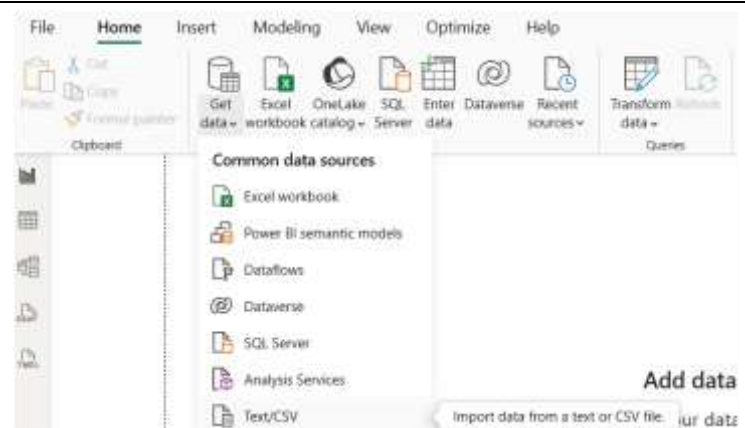
Data:

For this activity, The dataset contains information related to **sales transactions, customers, products, shipping details, and regional markets**. It includes tables such as **Sales, Customers, Products, Category, Subcategory, and Sales Executives**, allowing learners to practice **data modeling and create relationships between tables**. Using this dataset, learners can analyze **sales performance, profitability, customer segments, regional trends, and shipping costs** through different visualizations.

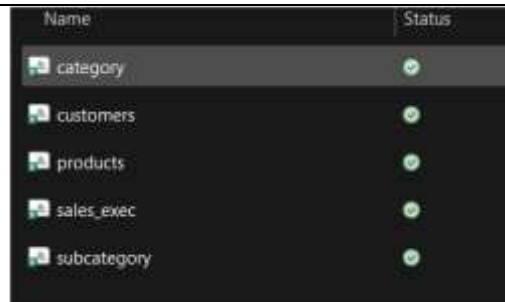
1. Launch **Power BI Desktop** on your system.
2. Once the application opens, you will see the **start screen** with multiple options to begin your work.
3. Under the section “**Select a data source or start with a blank report**”, click on **Blank report**.
4. This will open the **Power BI report canvas**, where you can start building your dashboard and importing data.



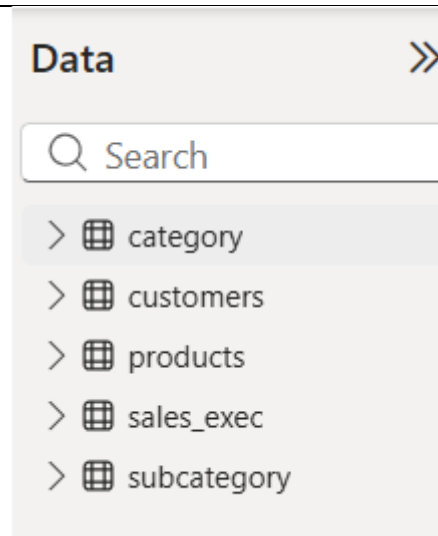
5. In the **Home** tab of Power BI Desktop, locate the **Get Data** option in the ribbon.
6. Click on the **Get Data** dropdown arrow to view available data source options.
7. Under **Common data sources**, you will see options such as:
 - Excel workbook
 - Power BI semantic models
 - Dataflows
 - Dataverse
 - SQL Server
 - Analysis Services
 - Text/CSV
8. Select the CSV source as we will import the data from CSV files.



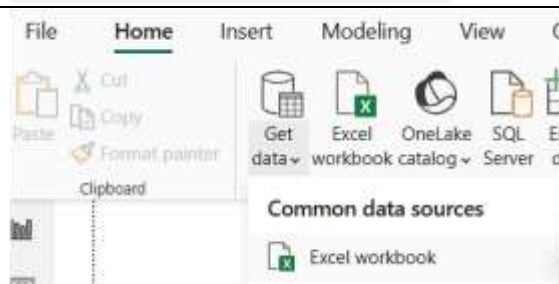
9. After selecting the **Excel workbook**, the **Navigator window** will appear.
10. In this window, Power BI displays all the **available tables or sheets** present in the Excel file.
11. Import the category table by clicking on it, and click on Load to import the table directly to Power BI desktop.
12. Repeat the previous step for all the remaining tables to import them one by one.



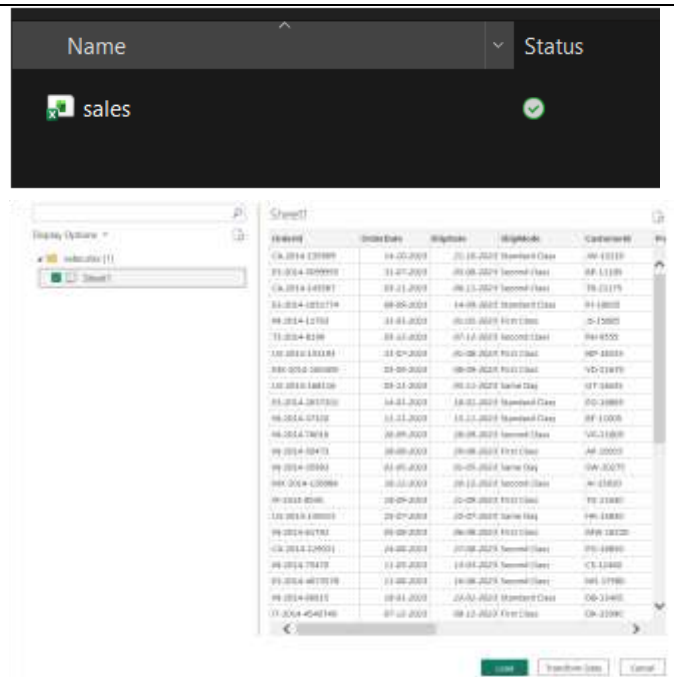
13. In the **Data (Fields) Pane** on the right side of Power BI, locate the tables that were imported.
14. Click the **arrow (>)** beside any table name to expand it.
15. Once expanded, you will be able to see the **individual columns (fields)** contained in that table.



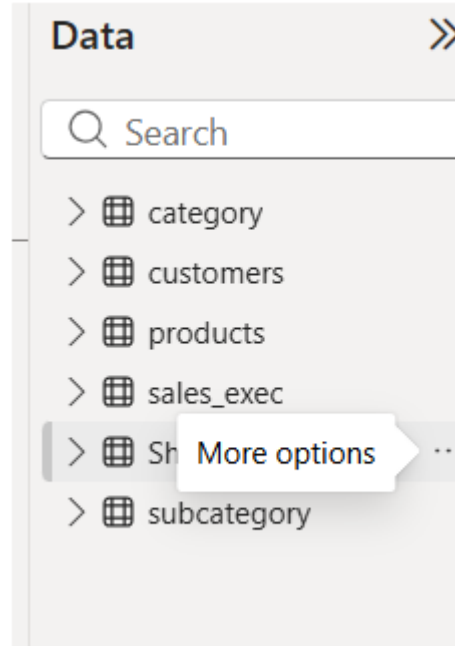
16. Go to the **Home** tab in the Power BI ribbon.
17. Click on the **Get Data** option.
18. From the list of **Common data sources**, select **Excel workbook**.
19. Browse to the location of the Excel file on your computer.

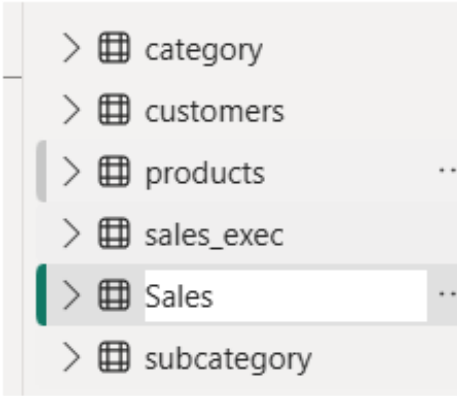
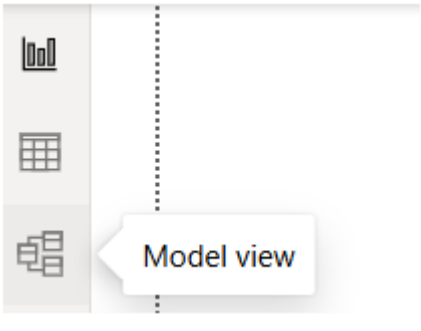
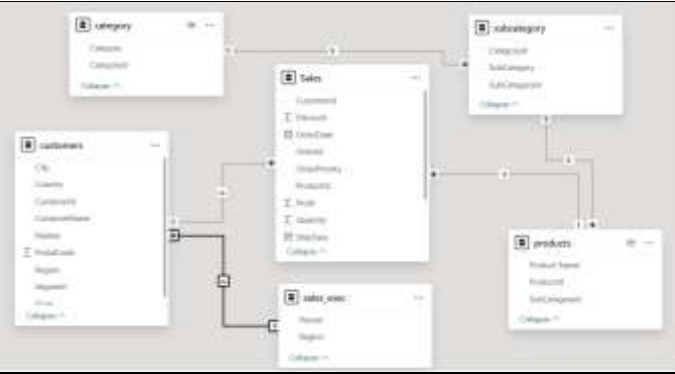


20. Select the **sales** table by checking the box next to it.
21. After selecting the **sales** table in the **Navigator window**, Power BI displays a **preview of the data**.
22. On the **left panel**, you can see the selected file and sheet (for example, **sales.xlsx → Sheet1**).
23. The **main preview area** shows the columns and rows from the dataset, such as:
 - OrderId
 - OrderDate
 - ShipDate
 - ShipMode
 - CustomerId
24. Review the data to confirm that the **correct sheet and structure** have been selected.
25. At the bottom of the window, click **Load** to import the data directly into Power BI.

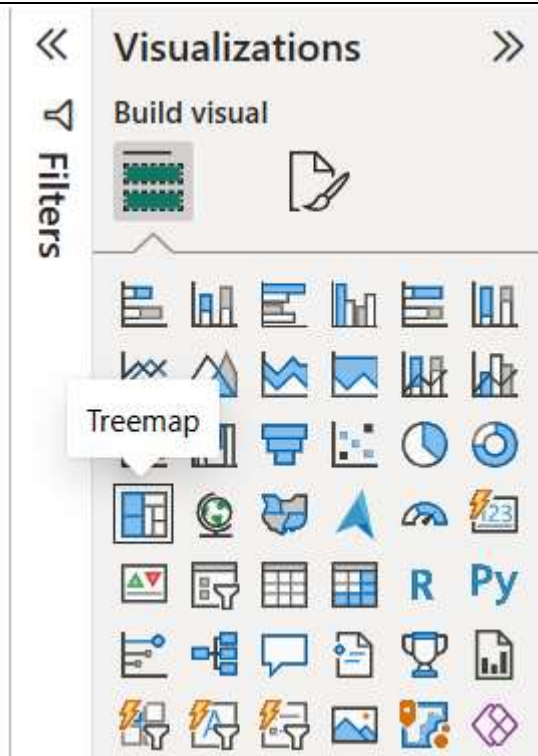


26. After loading the **sales dataset**, go to the **Data (Fields) pane** on the right side of Power BI.
27. Scroll through the list of available tables.
28. Locate the newly imported table named **Sheet1**.
29. Click the **three dots (More options)** beside the table and rename the table to **Sales**.

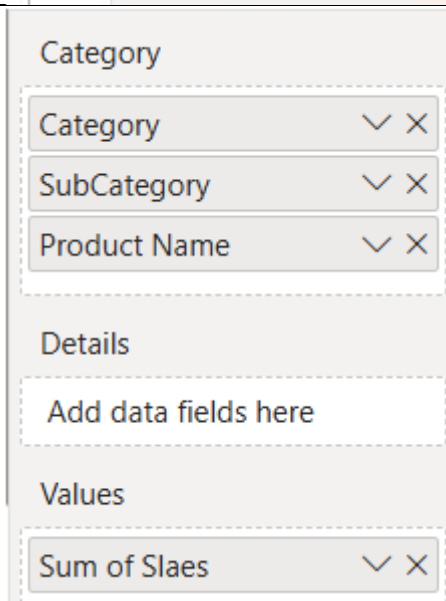


	
30. In Power BI Desktop, locate the left navigation panel. 31. Click on the Model View icon (the icon that looks like connected tables). 32. This will open the Model View workspace.	
33. In this view, Power BI displays all the tables in a relational diagram format. 34. You can use this view to create and manage relationships between tables.	

35. Go to the **Visualizations** pane on the right side of Power BI Desktop.
36. In the **Build visual** section, you will see multiple chart types available.
37. Locate the **Treemap visualization icon**.
38. Click on the **Treemap icon** to add a **Treemap visual** to the report canvas.
39. Once selected, a **blank Treemap visual** will appear on the report canvas where you can start adding fields.



40. After adding the **Treemap visual**, go to the **Visualizations** pane where the **field wells** appear.
41. Drag and drop the following fields from the **Data (Fields) pane** into the Treemap configuration areas.
42. **Category section**
 - Drag **Category** → into **Category**
 - Drag **SubCategory** → below **Category**
 - Drag **Product Name** → below **SubCategory**
43. This creates a **hierarchical structure** inside the Treemap.



-
- Drill Mode is on: Click a data point to drill
- Category
- Office Supplies
- Sum of Slaes by Category and SubCategory
- Phones
- Machines
- Copiers
- Accessories
- Sum of Slaes by Category, SubCategory and Product Name
- Apple Smart P... Nokia S... Noki... Appl... Sam... Se...
- Cisco Smart Ph... Samsung... Se...
- Motorola Sear... Cisco Sm... No...
- Nokia Smart P... Cisco Sm... Mo...
- Samsung Sear... Motorola... Ap...
- Product Name: Nokia Audio Dock, VoIP
- Sum of Slaes: 12,043.46

55. Each rectangle now represents **SubCategory sales contribution**.

56. For example, inside the **Technology category**, you may see subcategories such as:

- Phones
- Machines
- Copiers
- Accessories

57. With **Drill Mode still enabled**, click on any **SubCategory block**.

58. The Treemap will drill further down to display **individual products**.

59. Each rectangle now represents a **specific product** and its contribution to total sales.

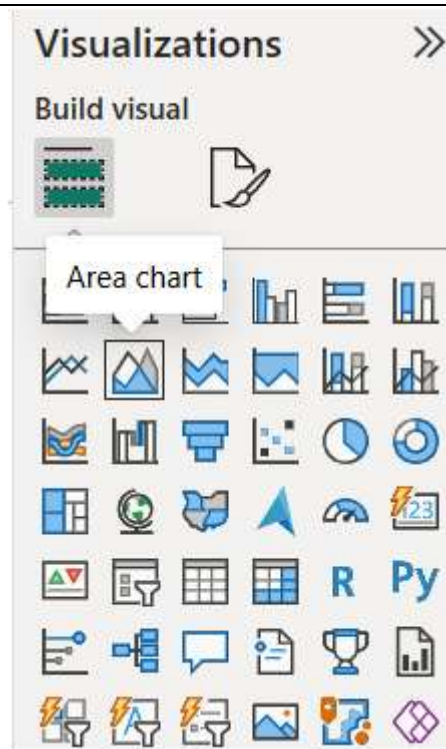
60. Hover over any product block to view details such as:

- Product Name
- Total Sales Value

Example:

*Nokia Audio Dock, VoIP →
Sales: 12,043.46*

61. Go to the **Visualizations** pane on the right side of the Power BI interface.
62. In the **Build visual** section, locate the **Area Chart** icon.
63. Click on the **Area Chart** visualization.



64. After inserting the Area Chart, go to the Visualizations pane where the field wells are displayed.
65. Drag and drop the following fields from the Data (Fields) pane into the corresponding areas.

X-axis

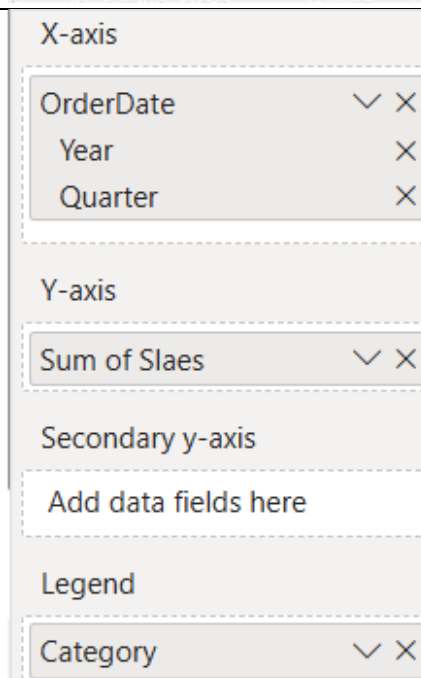
- Drag **OrderDate** into the **X-axis** field.
- From the date hierarchy, keep **Year** and **Quarter** levels.

Y-axis

- Drag **Sales** into the **Y-axis** field.
- Ensure the aggregation is **Sum of Sales**.

Legend

- Drag **Category** into the Legend field.



66. The Area Chart will now display Sales trends over time, separated by product categories.

67. After configuring the fields, the **Area Chart visualization** will appear on the **report canvas**.

68. The chart displays **Sales trends over time** based on the **Order Date hierarchy**.

69. The **X-axis** represents the **Year and Quarter**, showing the time progression.

70. The **Y-axis** represents the **Sum of Sales**.

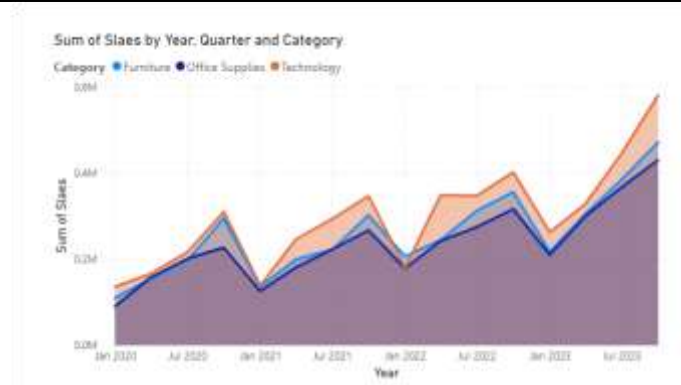
71. Different **colors represent different product categories**, such as:

- **Furniture**
- **Office Supplies**
- **Technology**

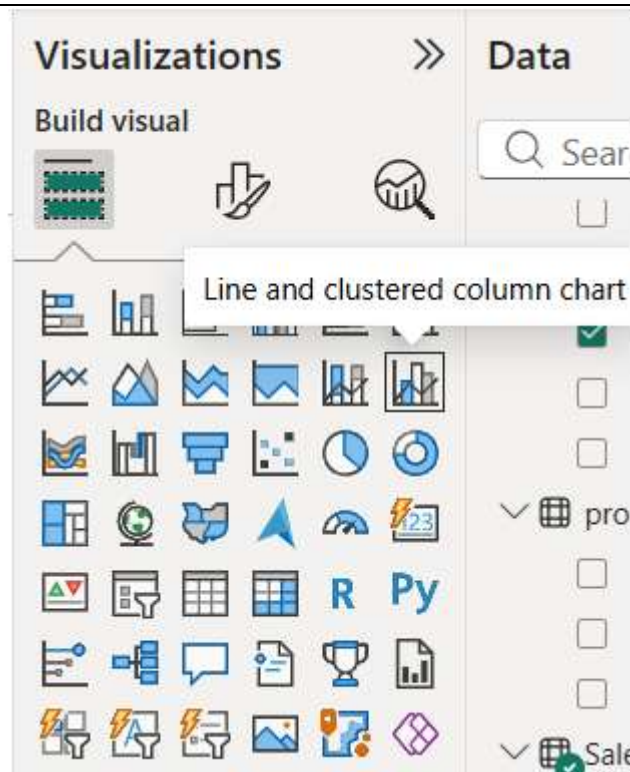
72. The filled areas make it easier to observe how **sales values grow or decline over time for each category**.

73. By examining the chart, you can identify:

- **Growth patterns** in different product categories
- **Periods of increased or decreased sales**
- **Which category dominates sales across the timeline**



74. Go to the **Visualizations** pane on the right side of the Power BI interface.
75. In the **Build visual** section, look for the **Line and Clustered Column Chart** icon.
76. Click the **Line and Clustered Column Chart** icon.



77. After adding the Line and Clustered Column Chart, go to the Visualizations pane where the field configuration section is displayed.

78. Drag and drop the following fields from the Data (Fields) pane into the respective areas.

X-axis

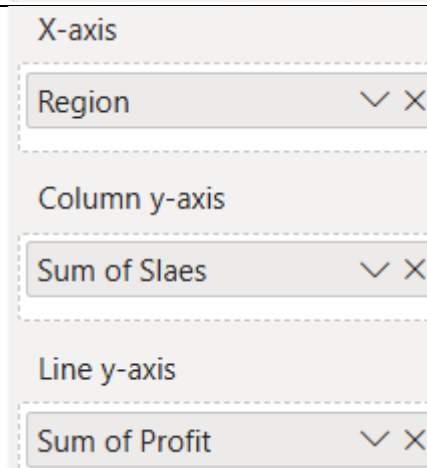
- Drag **Region** into the **X-axis** field.

Column y-axis

- Drag **Sales** into the **Column y-axis**.
- Ensure the aggregation shows **Sum of Sales**.

Line y-axis

- Drag **Profit** into the **Line y-axis**.
- Ensure the aggregation shows **Sum of Profit**.



79. Once the fields are assigned:
- Columns will represent Sales across different regions.
 - Line will represent Profit across the same regions.

80. Select the **Line and Clustered Column Chart** on the report canvas.

81. In the **Visualizations** pane, click the **Format Visual** icon (paintbrush icon).

82. The Format panel will open with multiple formatting options.

83. Expand the following sections to customize the visual:

X-axis

- Adjust axis titles, labels, and formatting if needed.

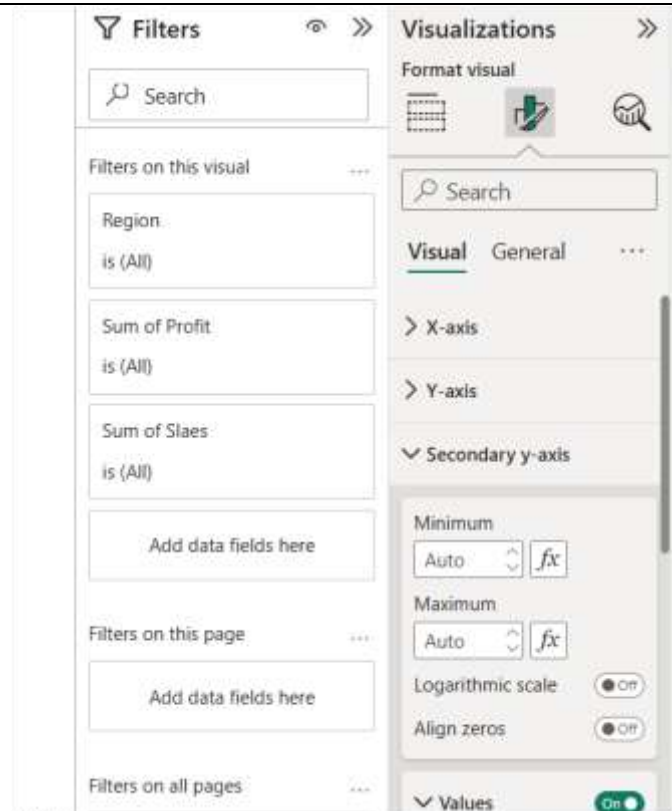
Y-axis

- Configure the scale and display settings for the **Sales values**.

Secondary Y-axis

- This axis represents the **Profit values (line chart)**.
- You can adjust settings such as:
 - **Minimum value**
 - **Maximum value**
 - **Logarithmic scale**
 - **Align zeros**

84. These options allow you to improve readability and

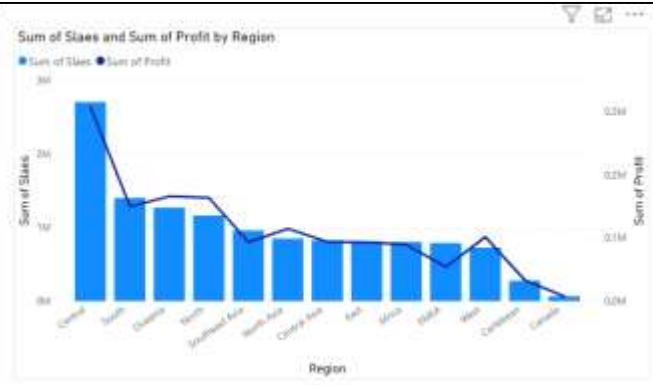


alignment between the Sales columns and Profit line.

85. The chart displays **two different measures simultaneously**:
- **Columns (Bars)** represent **Sum of Sales** for each region.
 - **Line** represents **Sum of Profit** for each region.

86. The **left Y-axis** represents **Sales values**, while the **right Y-axis** represents **Profit values**.

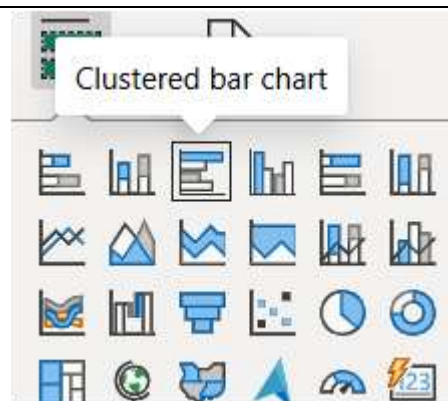
87. This visualization allows analysts to easily compare **regional sales performance with profitability trends**.
- By analyzing the chart, users can identify:
 - Regions with **high sales and high profit**
 - Regions with **high sales but relatively lower profit**
 - Regions with **low overall business contribution**



88. Go to the **Visualizations pane** on the right side of the Power BI interface.

89. In the **Build visual** section, locate the **Clustered Bar Chart icon**.

90. Click on the **Clustered Bar Chart** visualization.



91. In the Visualizations field configuration panel, drag and drop the following fields:

Y-axis

- Drag **Segment** into the **Y-axis** field.

X-axis

- Drag **Profit** into the **X-axis** field.
- Ensure the aggregation shows **Sum of Profit**.

Y-axis

Segment

X-axis

Sum of Profit

92. The chart will automatically update to display profit **distribution across different customer segments**.

93. The chart displays the **Sum of Profit across different customer segments**.

94. In this visualization:

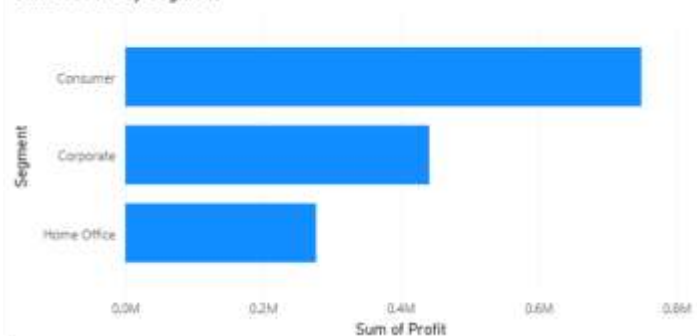
- The **Y-axis** represents the **Customer Segment**.
- The **X-axis** represents the **Sum of Profit**.

95. The segments displayed include:

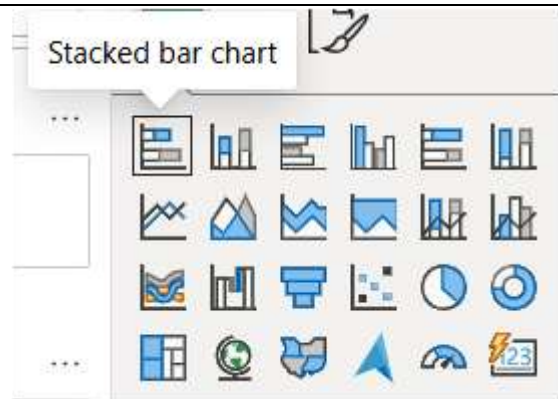
- Consumer
- Corporate
- Home Office

96. The **length of each bar represents the total profit generated by that segment**.

Sum of Profit by Segment



97. Go to the **Visualizations** pane on the right side of the Power BI interface.
98. In the **Build visual** section, locate the **Stacked Bar Chart** icon.
99. Click the **Stacked Bar Chart** option.



100. After inserting the Stacked Bar Chart, go to the Visualizations pane where the field configuration area is displayed.
101. Drag and drop the following fields from the Data (Fields) pane into the corresponding sections.

Y-axis

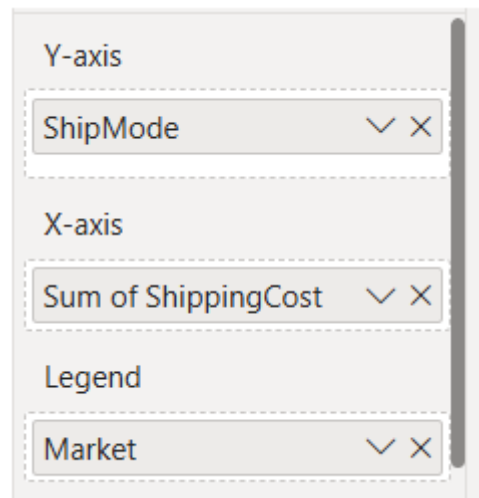
- Drag **ShipMode** into the **Y-axis** field.

X-axis

- Drag **ShippingCost** into the **X-axis** field.
- Ensure the aggregation shows **Sum of ShippingCost**.

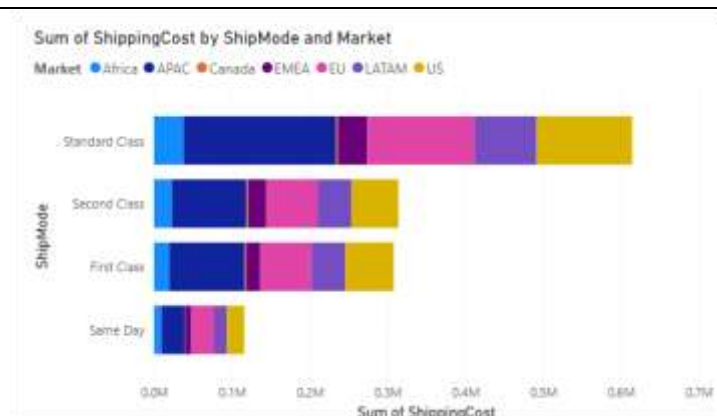
Legend

- Drag **Market** into the **Legend** field.



102. The chart displays the **Sum of Shipping Cost** grouped by **Ship Mode** and **Market**.

103. Each bar shows the **total shipping cost for a shipping mode**, while the colored segments represent **contributions from different markets**.



104. Go to the **Insert** tab in the Power BI ribbon.
105. Click on **Shapes** to view the available shape options.
106. From the dropdown, select **Rounded Rectangle**.
107. Draw the **rounded rectangle** on the report canvas by clicking and dragging your mouse.

- Select the **shape** you added to the report.
- In the **Visualizations** pane, open the **Format Shape** panel.
- Under the **Style** section, configure the following options:

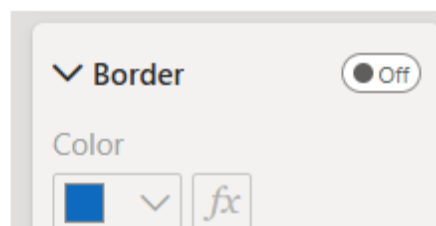
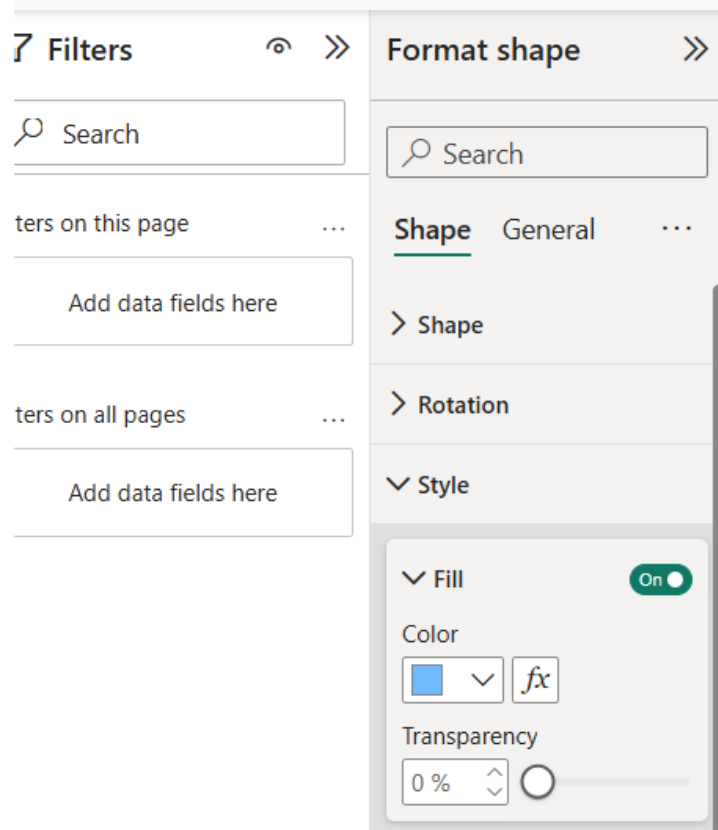
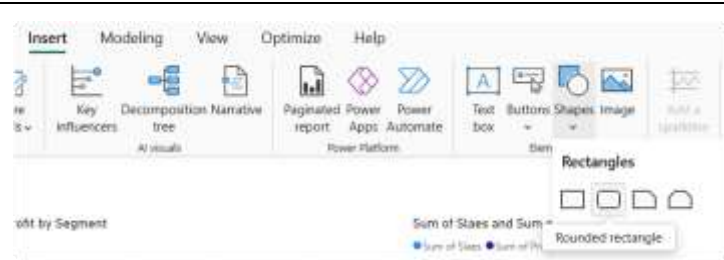
Fill

- Turn **Fill** → **On**.
- Choose a **color** that matches your preference.
- Adjust **Transparency** if needed.

Border

- Expand the **Border** section.
- You may **turn the border off** or customize its color and thickness.

108. Resize and position the shape so that it acts as a **background element or visual container** for charts.



109. Go to the Insert tab in the Power BI ribbon. 110. Click on Text box. 111. A text box will appear on the report canvas. 112. Click inside the text box and type the dashboard title. 113. Select the text box containing the title. 114. Adjust the formatting to make the title more prominent: • Increase the font size. • Apply bold formatting. • Choose a theme color for the text or background. 115. Resize the text box so it spans the top section of the report page. 116. Align the title to make it visually clear as the main heading of the dashboard.	